Exercise Sheet 3

Even though this PVK is structured into thematic sections and some content has not yet been covered, it is explicitly permitted and *should* be the case that in the problem sets *all the material* covered in the semester and in the lecture may be used.

Exercise 1
Differentiate the following functions.

$$(1)\ e^x\left(1-\sin^2(x)\right) \qquad (2)\ \sin\left(e^{x^2}\right) \qquad (3)\ \frac{e^x+1}{e^x-1} \qquad (4)\ x\cos(3^x)$$

Exercise 2 (FS2023 G. Felder)
Prove: Let f be a convex function $f:\mathbb{R}\to\mathbb{R}$, such that $f(x)\leq 2023$ for all $x\in\mathbb{R}$. Then f is constant.

Exercise 3
Let I be an interval and $f:I\to\mathbb{R}$ a differentiable function. Show that f is convex if and only if f' is monotonically increasing.

Exercise 4
Show that every continuous function $f:[a,b]\to\mathbb{R}$ is Riemann-integrable.

Exercise 5
Let (f_n) with $f_n:[a,b]\to\mathbb{R}$ be a sequence of integrable functions that converges uniformly to a function $f:[a,b]\to\mathbb{R}$. Prove, that f is integrable and

$$\int_a^b f(x)\,dx=\lim_{n\to\infty}\int_a^b f_n(x)\,dx.$$

Exercise 6 (FS2023 G. Felder)
Compute the limit of Riemann sums

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\sin\left(\frac{k}{n}\right).$$

Exercise 7
Let $f:[a,b]\to\mathbb{R}$ be a continuous function. Show that for every constant $C\in\mathbb{R}$ the function $F:[a,b]\to\mathbb{R}$ is an antiderivative of f and every antiderivative has the following form with a constant $C\in\mathbb{R}$:

$$F(x)=\int_a^x f(t)dt+C.$$

Exercise 8 (FS2024 A. Figalli)
Let $f:[0,1]\to\mathbb{R}$ be a continuously differentiable function.

(a) Suppose there exists an $x_0\in[0,1]$ such that $f(x_0)=0$. Show that

$$|f(x)|\leq\int_0^1|f'(s)|ds \qquad \forall x\in[0,1].$$

(b) Suppose that $\int_0^1 f(s)ds=0$. Show that

$$|f(x)| \leq \int_0^1 |f'(s)|ds \qquad \forall x \in [0,1]$$

and conclude that

$$\int_0^1 |f(s)|ds \leq \int_0^1 |f'(s)|ds.$$

(c) Let $m = \int_0^1 f(s)ds \in \mathbb{R}$. Show that

$$\int_0^1 |f(s) - m|ds \leq \int_0^1 |f'(s)|ds.$$